ROOT CAUSE UNLOCKED · CONDITION KIT

The Hashimoto's Root Cause *Kit*

A Root Health Condition Guide

Which thyroid numbers matter, what antibodies do and do not tell you, and the treatment question that is genuinely unsettled rather than simply under-discussed.

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The Hashimoto's Root Cause Kit

A Root Health Condition Guide

BEFORE YOU START

Contents

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. **Nothing here is a reason to change a thyroid medication dose on your own.** Levothyroxine has a narrow therapeutic window and self-adjusting it is genuinely risky.

Seek urgent care for chest pain, a resting heart rate persistently over 120, severe confusion or drowsiness, or a rapidly enlarging neck mass or difficulty swallowing or breathing. Thyroid disease has emergencies at both ends and they do not wait.

The drivers behind the antibodies

Unlock Your Hashimoto's works through the things that drive the autoimmune attack rather than only the hormone level. Testing exists to find out which of them are yours, so this guide follows the same order.

Driver	What it means	How you find it
The thyroid itself	How much hormone you are making and converting, and how hard the gland is being pushed	TSH, free T4, free T3, reverse T3
The autoimmune attack	Whether your immune system is targeting the gland, and how hard	TPO and thyroglobulin antibodies
The gut	Barrier integrity and microbial composition, the most studied non-thyroid driver	Celiac serology, stool analysis, permeability markers
The raw materials	Selenium, iron, vitamin D, zinc, B12. You cannot make or convert hormone without them	Blood work, covered below
The load	Chronic stress, poor sleep, and infectious burden acting on an already unstable immune system	History, and a small number of targeted tests

The thing to understand about this list: thyroid hormone replacement addresses the first row. It does nothing to the other four. That is not an argument against taking it, and for most people with Hashimoto's it is genuinely necessary. It is the reason people can be perfectly medicated and still feel unwell, and it is the whole rationale for this kit.

The thing most Hashimoto's material gets wrong

You have almost certainly been told your antibodies matter, and that lowering them is the goal.

Antibody titers are the most-measured, most-discussed, and least outcome-validated number in this condition. Several interventions genuinely lower them. **What nobody has demonstrated is that lowering them makes you feel better or changes where your disease goes.**

That is not a reason to ignore them. It is a reason to stop treating them as a scoreboard, and to be suspicious of anyone selling you a protocol whose success metric is a falling TPO number.

This guide tells you what each test actually establishes, and flags where the evidence stops.

The core panel

Marker	Unit	Standard lab range	Root Health functional target
TSH	mIU/L	0.4 to 4.0	1.0 to 2.5
Free T4	ng/dL	0.7 to 1.7	0.9 to 1.4
Free T3	pg/mL	2.0 to 4.2	2.8 to 3.6
Reverse T3	ng/dL	0 to 24	0 to 15
TPO antibodies	IU/mL	0 to 35	0 to 9
Thyroglobulin antibodies	IU/mL	0 to 20	0 to 1
TSI / TRAb	%	0 to 140	0 to 99
Ferritin	ng/mL	25 to 300	50 to 150
Vitamin B12	pg/mL	200 to 1100	500 to 900
Folate (RBC)	ng/mL	400 to 2000	600 to 1500
Zinc (plasma)	ug/dL	60 to 120	70 to 110
Selenium (serum)	ug/L	60 to 160	85 to 140
hs-CRP	mg/L	0 to 1.0	0 to 0.5

These are clinical targets used to guide investigation. **They are not validated diagnostic thresholds**, and a result outside a functional window is not a diagnosis of anything.

What each one is actually for.

TSH is the pituitary's opinion of whether there is enough thyroid hormone, not a direct measure of thyroid hormone. It is the most sensitive single screening test and it is also the one most affected by time of day, recent illness, and biotin supplements. Stop biotin for several days before testing, because it interferes with the assay and can produce results that look like thyroid disease when there is none.

Free T4 and Free T3 measure the hormones themselves. Free T4 is the storage form, Free T3 the active form.

Reverse T3 is worth a specific caution. It is elevated by illness, calorie restriction, and stress, which are exactly the situations where the body deliberately downshifts. A high reverse T3 usually reflects that the body is doing something appropriate in response to a stressor, and the useful question is what the stressor is, not how to lower the number.

TPO and thyroglobulin antibodies establish that the process is autoimmune. See the next section for what they do not do.

Ferritin, B12, folate, zinc, selenium are here because their deficiencies produce fatigue, hair loss, and cognitive fog that get attributed to thyroid disease and then persist after the thyroid is well managed. Low ferritin in particular is extraordinarily common in this population and is the single most frequent reason someone with a perfect TSH still feels terrible.

What antibodies do and do not tell you

What they establish: that your hypothyroidism is autoimmune rather than something else, which is diagnostically useful. Positive antibodies with a normal TSH also predict a higher likelihood of developing hypothyroidism later, which justifies monitoring rather than action.

What they do not establish: how you feel, how fast the disease will progress, or whether a treatment is working.

Two things reliably lower them, and here is the honest read on both.

Vitamin D. A meta-analysis of 12 studies covering 862 people found vitamin D supplementation significantly reduced TPO antibody titers, with a standardized mean difference of 1.084, and thyroglobulin antibodies at 0.996.[1] Those are large effects on the antibody number.

Selenium. An umbrella review of 6 systematic reviews covering 75 randomized trials found TPO antibodies fell significantly at 3 months in levothyroxine-treated patients taking selenium, with a standardized mean difference of 0.53. **Only one of the six reviews was rated high quality.**[2]

And here is what a Cochrane review found when it asked the harder question. Four randomized trials, 463 participants, all at unclear to high risk of bias. **Change in health-related quality of life was not assessed in a single one of them.** Neither was change in levothyroxine dose requirement. The authors concluded the evidence does not allow confident decision making about selenium in Hashimoto's.[3]

So: the antibody number moves. Whether the person moves with it has largely not been measured. That is the most important sentence in this guide.

The treatment question that is genuinely unsettled

If your TSH is elevated but your free T4 is normal, that is subclinical hypothyroidism, and whether to treat it is a real controversy rather than a settled question your doctor is getting wrong.

A JAMA systematic review and meta-analysis of 21 publications randomizing 2,192 adults found that thyroid hormone therapy lowered TSH into the normal range compared with placebo, and examined the effect on quality of life and thyroid-related symptoms.[4]

A 2026 systematic review in older adults is titled, in its own words, “no evidence of benefit,” evaluating levothyroxine against placebo or no treatment on health-related quality of life.[5]

But it is not one-sided. A meta-analysis of 13 studies in older patients with subclinical hypothyroidism found levothyroxine significantly reduced total cholesterol, triglycerides, LDL cholesterol, and apolipoprotein B.[6] So there may be a metabolic argument even where there is not a symptomatic one.

What to do with that. If you have been told your subclinical hypothyroidism does not need treating, your doctor is standing on real evidence, not dismissing you. If you have been treated and feel no different, that is also the expected finding rather than a failure on your part. This is a legitimate judgment call, and it is worth having as a conversation rather than a fight.

Reading your results as a pattern

High TSH, low free T4, positive antibodies. Overt autoimmune hypothyroidism. Treatment is not controversial here.

High TSH, normal free T4, positive antibodies. Subclinical, autoimmune. See the section above. Worth monitoring regardless.

Normal TSH, positive antibodies, feeling unwell. Common, and frustrating. The antibodies confirm autoimmunity but do not explain current symptoms. This is where the ferritin, B12, folate, zinc, and vitamin D rows earn their place, along with sleep and the rule-outs below.

Normal everything, feeling unwell. Thyroid has been investigated. The search should continue, not stop.

Normal TSH on treatment, still symptomatic. The most common scenario in the whole condition. Check ferritin first. Then B12 and vitamin D. Then sleep apnea, which is routinely missed in women and produces exactly this picture.

What else produces this picture

Worth ruling out	Why it mimics
Iron deficiency, with or without anemia	Fatigue, hair loss, cold intolerance, poor exercise tolerance
Vitamin B12 deficiency	Fog, fatigue, nerve symptoms
Vitamin D deficiency	Diffuse aching, fatigue, low mood
Obstructive sleep apnea	Unrefreshing sleep, fatigue, fog. Missed constantly in women
Celiac disease	Coexists with Hashimoto's more often than chance, and is a real diagnosis with real testing
Perimenopause	Overlaps almost completely in the age group where Hashimoto's presents
Depression	Fatigue, cognitive difficulty, sleep disruption
Anemia of any cause	Fatigue, breathlessness on exertion

Celiac deserves a note. It genuinely coexists with autoimmune thyroid disease more often than chance would predict. **Get tested while still eating gluten.** Going gluten free first makes the testing unreliable and costs you a clean answer. The diet guide in this kit covers what a head-to-head trial actually found about gluten-free diets in Hashimoto's, and it is not what most thyroid content will tell you.

What to ask your clinician

- "Can we check ferritin, B12, vitamin D, and zinc alongside the thyroid panel? I want to rule out the things that feel like thyroid."
- "My TSH is in range but I still feel unwell. What else is worth looking at?"
- "Have I been screened for celiac disease, and do I need to be eating gluten for that test to be valid?"
- "Given my TSH and free T4, what is your view on treating versus monitoring, and what would change your mind?"
- "Should I be screened for sleep apnea?"

Bring: your full result history rather than the latest one, since the trend matters more than any single draw; the time of day you were tested; your full medication and supplement list including biotin; and a symptom timeline.

Testing consistency matters more than people realize. Same lab, same time of day, same relationship to your medication dose. TSH varies through the day, and comparing a morning draw to an afternoon one produces changes that are not real.

When to seek clinical review

Urgently: chest pain, resting heart rate persistently over 120, severe confusion or drowsiness, rapidly enlarging neck mass, or difficulty swallowing or breathing.

Promptly: new or rapidly worsening symptoms after a dose change, or symptoms of over-replacement including palpitations, tremor, heat intolerance, and unintended weight loss.

Before going gluten free: if celiac is a question, test first.

If you are pregnant or planning pregnancy: thyroid management changes substantially and requirements usually rise. This needs a clinician, promptly, and it is one of the few areas where the evidence for treating subclinical hypothyroidism is stronger.

Track what changes

TSH is checked every few months. How you actually feel changes week to week, and nobody records it, so the conversation defaults to the number.

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Consultations are available to people located in North Carolina, where I am licensed. If you are elsewhere, the request list above works in any office.

The gut, which is the most studied driver after the gland itself

Key 3 of the ebook is the gut-thyroid connection, and this is where the research has moved fastest.

What the evidence shows. A systematic review and meta-analysis found altered gut microbiota composition in autoimmune thyroid disease at the family, genus, and species levels compared with controls (PMID 34867823). A separate study characterizing the microbiome in primary hypothyroidism concluded that hypothyroidism causes changes in the gut microbiome and that, in mice, an altered flora can in turn affect thyroid function, which is what makes it an axis rather than a one-way street (PMID 32519746).

A 2026 review of the gut-thyroid axis describes dysbiosis, impaired intestinal barrier function, and altered microbial metabolites, particularly short-chain fatty acids, as contributors to immune imbalance, and calls the gut microbiota **a modifiable contributor to thyroid autoimmunity** while naming the honest gap: causality still needs clarifying (PMID 41292506).

Read that carefully. “Modifiable contributor” with causality unresolved is a fair reason to work on the gut. It is not a promise that fixing the gut resolves the antibodies.

What to ask for. Celiac serology first, drawn while you are still eating gluten. Then, if gut symptoms are present, a comprehensive stool analysis. The companion Leaky Gut kit covers barrier markers in more depth, and the caution there applies here too: read a barrier panel as a pattern, not as proof from any single marker.

The load: stress, sleep, and infection

This section was missing from the first version of this kit, and it is a genuine part of the picture rather than a wellness postscript.

Stress, with the citation limits stated plainly. The controlled human work linking stressful life events to autoimmune thyroid disease is in **Graves' disease**, not Hashimoto's. One controlled study found an excess of life events in the year before Graves' onset (PMID 8498147), and another found that patients with Graves' reported more negative stressful life events than a comparison group with toxic nodular goitre, suggesting life events act as a precipitating factor (PMID 11453947).

I am telling you the disease those studies were done in, because most sources citing them do not. The honest position is that stress is an established precipitant in one autoimmune thyroid disease and a reasonable clinical concern in the other, not a proven cause of Hashimoto's.

What follows from that. A daily down-regulating practice, protected sleep, and a realistic look at load are worth doing on their own terms. They are free, they carry no interaction risk, and the mechanism they act on, immune regulation, is the same one the antibodies live in.

- Five to ten minutes daily of slow breathing with a longer exhale than inhale, done consistently.
- A fixed sleep window of seven to nine hours. Hypothyroid fatigue and poor sleep hygiene compound each other, and only one of them is treatable with a tablet.
- Movement you can sustain. Not high-intensity work on a bad fatigue day, which reliably backfires.

Infection, as a plausible driver rather than a certainty. Epstein-Barr virus has drawn sustained attention for a potential role across thyroid disease including Hashimoto's, and a 2025 review integrates the immunological and virological evidence (PMID 41229422). Note what that is: a mechanistic and associational review, not a trial showing that treating a virus changes thyroid antibodies. There is no validated antiviral protocol for Hashimoto's, and anyone selling you one is ahead of the evidence.

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Functional target ranges transcribed from the Root Health blood chemistry interpreter.

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This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing here is a reason to change a thyroid medication dose on your own.

This guide is going to disagree with the most common piece of Hashimoto’s diet advice on the internet. I want to be clear up front that I am not dismissing gluten. I am telling you what the evidence shows and why the standard advice has the order of operations backwards.

Start here: the medication timing detail

Before any diet change, this one is worth more to most readers than everything else in the guide, and it takes no willpower at all.

Levothyroxine absorption is reduced by things people take at breakfast. The prescribing information for levothyroxine lists interference from calcium supplements, iron supplements, and other minerals, and absorption is affected by food generally. Coffee and soy are also commonly implicated.

What to do: take it on an empty stomach with water, and separate it from coffee, food, calcium, iron, magnesium, and multivitamins. Most guidance suggests 30 to 60 minutes before eating, and separating minerals by around four hours.

Why this matters more than it sounds. If your absorption is inconsistent, your dose is inconsistent, and no diet change will show through that noise. People spend months on elimination diets while taking their thyroid medication with a mineral-fortified breakfast. Fix the timing first, hold it steady for six weeks, and then judge everything else against a stable baseline.

Do not change your dose to compensate. Change the timing, tell your prescriber you have done so, and retest when they suggest.

The gluten question, and why the usual advice is backwards

The standard recommendation is: you have Hashimoto's, go gluten free. Two separate lines of evidence say that is the wrong first move.

A head-to-head comparison. A pilot study compared the Mediterranean diet, a gluten-free diet, and a free diet in people with Hashimoto's, measuring oxidative stress markers over 12 weeks.[1]

The Mediterranean group showed significantly lower advanced glycation end products and significantly higher glutathione peroxidase, thioredoxin, and total antioxidant capacity, both compared with their own baseline and compared with the other two groups. **In the gluten-free group, the oxidative stress parameters did not change significantly.**

And a finding that applies to all three arms: **no significant change in TSH, free T4, thyroglobulin antibodies, or TPO antibodies in any group.** No diet moved thyroid function or thyroid antibodies in twelve weeks.

The authors' own conclusion is that a gluten-free diet does not significantly influence markers of oxidative stress or thyroid autoimmunity and function. It is a pilot study and should be read as one.

A genetic causality analysis. A Mendelian randomisation study examined whether the relationships here are causal.[2] Two findings.

Hashimoto's disease **increases the risk of celiac disease**, with an odds ratio of 1.544 and a confidence interval of 1.153 to 2.068. That association is real and it runs in that direction.

And the study "does not support genetic liability related to GFD with AITD," concluding that **a gluten-free diet has no protective effect on autoimmune thyroid disease** and that any benefit people experience may work through other mechanisms.

A 2026 review states the position well: in celiac disease the pathogenic role of gluten is firmly established, but in Hashimoto's, gluten is better viewed as a **candidate modifier rather than a proven universal trigger**.[3]

So what should you actually do

Get tested for celiac disease, while still eating gluten. This is the part the standard advice skips and it is the part that matters. Hashimoto's raises your celiac risk measurably. Celiac is a real diagnosis with real testing and a treatment that genuinely works, and the testing depends on you currently eating gluten. Go gluten free first and you lose the ability to get a clean answer without a deliberate gluten challenge later.

If celiac testing is positive, gluten free is the treatment. Not a wellness experiment, a treatment, and it matters enormously.

If celiac testing is negative and you still want to try it, that is reasonable. Run it as a time-limited experiment with a defined endpoint, judged on how you feel rather than on a repeat antibody panel, since the antibody panel did not move in the trial. Give it eight to twelve weeks and then decide honestly.

What I would not do is adopt a permanent restriction, on the strength of evidence that does not support it, in a condition where the same trial showed a Mediterranean pattern outperforming it.

The pattern that did outperform

In the only head-to-head in this guide, the Mediterranean pattern was the one that moved markers.[1] It is also less restrictive, cheaper, easier to sustain, and better supported for general health than any elimination diet.

Build meals from:

Vegetables in volume and variety, across colors. Olive oil as the main fat. Fish, particularly oily fish, a couple of times a week. Legumes. Whole grains if you tolerate them and celiac has been excluded. Nuts and seeds. Fruit, with an emphasis on berries. Herbs and spices generously.

Moderate: red meat, and processed meat more so. Refined carbohydrate and added sugar. Alcohol.

This is not a thyroid intervention and I am not going to sell it as one. No diet in that trial changed thyroid function or antibodies. What the Mediterranean arm changed was oxidative stress markers, which is a mechanistic outcome rather than a symptom one. It is a good way to eat with the best available support in this specific condition. That is the honest size of the claim.

Iodine, where more is not better

This one matters because the advice circulating is often actively harmful.

Iodine is required to make thyroid hormone, which leads people to conclude that more iodine means better thyroid function. In autoimmune thyroid disease that reasoning breaks down.

Excess iodine is implicated as a risk factor in Hashimoto's. In a study of risk factors for autoimmune thyroid disease, median urinary iodine was significantly higher in the autoimmune group than in controls.[4] Mechanistic work found that excessive iodine activates a hypoxia pathway promoting apoptosis of thyroid follicular cells, described by the authors as potentially an important risk factor contributing to Hashimoto's development.[5]

What this means practically. Do not take high-dose iodine or kelp supplements for Hashimoto's without a specific reason and supervision. Kelp in particular can deliver very large and very variable doses. Normal dietary iodine from a mixed diet, including iodized salt, is a different matter and is not the concern here.

If you are pregnant or planning pregnancy, iodine requirements rise and this becomes a conversation with your clinician rather than a rule from a guide.

Cruciferous vegetables, which are probably fine

Broccoli, cabbage, kale, and Brussels sprouts contain compounds that can interfere with iodine uptake, which is why they carry a reputation in thyroid circles.

The practical reality is that the amounts required to affect thyroid function are far beyond normal consumption, cooking reduces the effect substantially, and these are among the more nutritious foods available. In a population where iodine excess is the more relevant concern than iodine deficiency, avoiding them is solving the wrong problem.

Eat them. If you are eating raw kale in very large daily quantities and have low iodine intake, that is worth mentioning to your clinician. Otherwise this is a worry you can put down.

The nutrients that actually explain how you feel

The lab guide in this kit covers testing. From a food standpoint:

Iron is the big one. Low ferritin is extremely common in this population and produces fatigue, hair loss, and cold intolerance that get attributed to the thyroid and then persist after the thyroid is managed. Red meat, poultry, fish, legumes, and dark leafy greens, with vitamin C at the same meal to aid absorption from plant sources. **Take iron supplements away from levothyroxine.**

Selenium is found in seafood, eggs, and Brazil nuts. Brazil nuts are extremely concentrated and vary enormously nut to nut, so a couple a day is plenty and a handful is too many.

Zinc from shellfish, meat, legumes, and seeds.

Vitamin D is difficult to get from food in meaningful quantities. Testing beats guessing.

B12 from animal foods, with the caveat from the lab guide that certain medications interfere with absorption.

Running a fair experiment

Fix the medication timing first, hold it six weeks, and let a stable baseline establish itself. Everything else is measured against that.

Then change one thing. Mediterranean pattern first, since it is the one with head-to-head support and it costs you the least.

Give it eight to twelve weeks. The trial ran twelve.

Do not judge it on antibodies. No diet arm in the trial moved them. Judge it on energy, sleep, cognition, and how you actually feel, recorded rather than remembered.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=hashimotos-diet

Realistic expectations

No diet in the available head-to-head changed thyroid function or thyroid antibodies.[1] A gluten-free diet has no demonstrated protective effect on autoimmune thyroid disease.[2] Gluten is a candidate modifier, not a proven universal trigger.[3]

What diet realistically offers here is a better background: lower oxidative stress on a Mediterranean pattern, better nutrient status for the deficiencies that masquerade as thyroid symptoms, and the avoidance of two specific mistakes, excess iodine and poorly timed medication.

That is worth doing. It is not a cure and it will not replace treatment.

And the most valuable single action in this guide remains the celiac test, taken while still eating gluten, because that is the one path where a dietary change is genuinely the treatment.

When to seek clinical review

Urgently: chest pain, resting heart rate persistently over 120, severe confusion or drowsiness, or difficulty swallowing or breathing.

Before going gluten free: get celiac testing while still eating gluten.

Before taking iodine or kelp: ask, particularly with autoimmune thyroid disease.

If you change medication timing: tell your prescriber, because absorption may improve and your dose may need review.

If pregnant or planning pregnancy: thyroid and iodine requirements both change and this needs clinical management.

The gut connection behind the diet

Diet is one input into the gut-thyroid axis, and it helps to know what that axis is. A systematic review and meta-analysis found altered gut microbiota composition in autoimmune thyroid disease compared with controls (PMID 34867823), and a 2026 review describes dysbiosis, impaired barrier function, and altered short-chain fatty acid production as contributors to immune imbalance, calling the microbiota a modifiable contributor while noting that causality is still being worked out (PMID 41292506).

Practically that means fiber diversity and fermented foods have a rationale here beyond general health advice, and it also means no diet has been shown to reverse thyroid autoimmunity. Both things are true.

Working with Root Health

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Consultations are available to people located in North Carolina, where I am licensed.

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Levothyroxine absorption guidance reflects the product prescribing information rather than a specific trial.

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Every supplement in this guide has to be timed around levothyroxine. That is not a footnote, it is the most consequential practical point in the document, and it has its own section.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, "no evidence" has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed*: a real trial ran and the answer was no. *Never properly tested*: nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won*: real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

The load: stress, sleep, and infection

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What follows from that. A daily down-regulating practice, protected sleep, and a realistic look at load are worth doing on their own terms. They are free, they carry no interaction risk, and the mechanism they act on, immune regulation, is the same one the antibodies live in.

- Five to ten minutes daily of slow breathing with a longer exhale than inhale, done consistently.
- A fixed sleep window of seven to nine hours. Hypothyroid fatigue and poor sleep hygiene compound each other, and only one of them is treatable with a tablet.
- Movement you can sustain. Not high-intensity work on a bad fatigue day, which reliably backfires.

Infection, as a plausible driver rather than a certainty. Epstein-Barr virus has drawn sustained attention for a potential role across thyroid disease including Hashimoto's, and a 2025 review integrates the immunological and virological evidence (PMID 41229422). Note what that is: a mechanistic and associational review, not a trial showing that treating a virus changes thyroid antibodies. There is no validated antiviral protocol for Hashimoto's, and anyone selling you one is ahead of the evidence.

The finding this guide is built around

Two studies, read side by side, tell you something no single one of them says.

The intervention that most improved quality of life did not change antibodies at all. In a double-blind randomised study, women with Hashimoto's received nutritional education plus either a probiotic (*Lactiplantibacillus plantarum* 299v) or placebo. The probiotic group improved overall quality of life far more, 60.94 points versus 35.94, and improved in 12 of 14 domains versus 3 of 14.[1]

And: anti-TPO levels did not improve in either group. The authors conclude the benefits "appear to be independent of anti-TPO levels."

Meanwhile the interventions that reliably lower antibodies have never been shown to make anyone feel better. The Cochrane review of selenium found that health-related quality of life was not assessed in a single one of the four trials.

Put those together and you get the operating principle for this whole guide: **the antibody number and how you feel are two different outcomes, and the things that move one are not necessarily the things that move the other.** Most Hashimoto's supplement advice optimizes the number. This guide tries to optimize you.

Timing, before anything else

Levothyroxine absorption is reduced by minerals and by food. The prescribing information lists interference from calcium and iron supplements in particular.

The rules:

- Levothyroxine on an empty stomach with water, 30 to 60 minutes before eating
- **Calcium, iron, magnesium, zinc, and multivitamins separated by around four hours**
- Coffee separated from the dose
- Consistency matters more than perfection. Same routine daily beats an optimal routine done erratically

Why this outranks every supplement below. If your absorption swings, your effective dose swings, and nothing you add will be interpretable through that noise. Fix timing, hold it six weeks, then start experimenting.

If you change your timing, tell your prescriber. Absorption may improve, which can mean you are now on a higher effective dose than intended.

Tier 1: The one with the most support

Selenium

What the evidence shows, in both directions.

A network meta-analysis comparing different supplements in Hashimoto's concluded, after examining the alternatives, that patients should be given an appropriate amount of selenium as an auxiliary treatment alongside standard care.[2] An umbrella review of 6 systematic reviews covering 75 trials found TPO antibodies fell significantly at 3 months in levothyroxine-treated patients on selenium, standardized mean difference 0.53, though only one of the six reviews rated high quality.[3]

And the Cochrane review, which asked the harder question. Four trials, 463 participants, all at unclear to high risk of bias. **Change in quality of life was not measured in any of them.** Neither was change in levothyroxine dose requirement. The authors concluded the evidence does not permit confident decision making.[4]

My honest read. Selenium has the best antibody evidence of anything here and the weakest evidence that it changes your life. It is reasonable to trial, it is inexpensive, and you should not expect to feel it.

Commonly studied amount: 200 mcg daily in the trials. **Functional serum target used at Root Health:** 85 to 140 ug/L. This is one worth testing rather than assuming. **Cautions, and these matter:** selenium has a genuinely narrow window. Chronic high intake causes selenosis, with hair and nail changes, garlic breath, and gastrointestinal upset. **Do not stack multiple products containing selenium**, and remember Brazil nuts are extremely concentrated and vary enormously nut to nut. A couple a day is plenty; a handful is too many. This is not a more-is-better nutrient.

Tier 2: Depends on your situation

Iron, if your ferritin is low

Not a thyroid treatment. It is on this list because **low ferritin is the single most common reason someone with a well-controlled TSH still feels terrible**, and because the symptoms overlap almost completely: fatigue, hair loss, cold intolerance, poor exercise tolerance.

Functional ferritin target used at Root Health: 50 to 150 ng/mL, against a standard range of 25 to 300. A great many people sit between 25 and 50, get told it is normal, and are symptomatic.

Cautions: test before supplementing, because iron overload is a real condition and iron is not a benign supplement. **Separate from levothyroxine by four hours.** Take with vitamin C, away from calcium and coffee.

Vitamin D, where two meta-analyses disagree

I am going to show you the disagreement rather than pick the answer I prefer.

One meta-analysis of 12 studies covering 862 people found vitamin D significantly reduced TPO antibodies, standardized mean difference 1.084 with a confidence interval of 1.624 to 0.545, and thyroglobulin antibodies at 0.996.[5]

A network meta-analysis found vitamin D did not effectively reduce TPO antibodies, with a standardized mean difference of 2.54 and a confidence interval running from 6.51 to positive 1.42, which crosses zero.[2]

Different methods, different included trials, opposite conclusions. When two meta-analyses of the same question disagree this cleanly, the honest summary is that the answer is not settled.

What I would do anyway: correct a documented deficiency, because that is defensible on general grounds regardless of what it does to antibodies. Do not take it as an antibody treatment.

Functional target: 40 to 80 ng/mL. Test rather than guess. Fat soluble and genuinely possible to overdo.

A probiotic, for quality of life rather than for antibodies

Based on the study that opens this guide.[1] *Lactiplantibacillus plantarum* 299v, added to nutritional change, improved quality of life substantially more than nutritional change alone, across 12 of 14 domains.

Be precise about the claim. It did not change anti-TPO. It did not change BMI. It changed how people reported feeling, in a double-blind randomised design. That is a legitimate and under-valued outcome, and it is the specific thing most Hashimoto's supplement protocols never measure.

Caution: this is one strain in one study. Do not generalize it to probiotics as a category.

Zinc

Included because deficiency is common and produces overlapping symptoms, and because it is on the functional panel. Functional target 70 to 110 ug/dL. **Separate from levothyroxine by four hours.** Zinc taken long term can drive copper deficiency, so this is not one to take indefinitely without a reason.

Tier 3: Ask your practitioner first

No products and no purchase links here, on purpose. These are things worth raising with a clinician, and selling them from a page to someone I have never assessed would contradict the rest of this guide.

Myo-inositol combined with selenium. A 2026 systematic review with trial sequential analysis found myo-inositol plus selenium significantly reduced TSH compared with selenium alone in Hashimoto's with subclinical hypothyroidism, but found **no significant difference in T4 or in TPO antibodies**.[6] The authors urge caution given the limited number of studies. A TSH-lowering supplement is a real thing to discuss with whoever manages your dose, not something to add quietly. *Ask:* "This lowered TSH in trials. Given I am on levothyroxine, does adding it risk over-replacement?"

T3 or desiccated thyroid, if you feel unwell on levothyroxine alone. This comes up constantly and deserves a straight answer. A meta-analysis of blinded randomized trials found that **approximately half of participants preferred combination T3 and T4 therapy over T4 alone, which was not distinguishable from chance.**[7] A separate meta-analysis of 16 studies examined combination therapy and desiccated thyroid directly.[8] So: some people genuinely do better, the trials cannot yet tell us who, and it is a legitimate conversation rather than either a miracle or a fringe request. *Ask:* "I am biochemically controlled and still symptomatic. What is your view on a trial of combination therapy, and what would we measure?"

Anything at all, if you are pregnant or planning pregnancy. Requirements change substantially, iodine becomes a live question, and thyroid management in pregnancy is one of the few areas where treating subclinical hypothyroidism has stronger support.

High-dose iodine or kelp. Covered in the diet guide. Excess iodine is implicated as a risk factor in this specific condition, which makes it the one supplement where the popular advice is pointed the wrong way.

What I deliberately left out, and why

Anything sold as reversing Hashimoto's. Nothing has demonstrated that.

Antibody titers as a success metric. The probiotic study improved quality of life across 12 of 14 domains without moving anti-TPO.[1] The selenium trials moved anti-TPO without ever measuring quality of life.[4] A protocol that judges itself on your antibody number has chosen the outcome that is easier to move, not the one you came for.

Proprietary thyroid support blends. They routinely contain iodine, often a lot, sometimes undisclosed. In autoimmune thyroid disease that is the wrong direction, and an undisclosed amount means you cannot check.

Things I use in practice that are not in here. A guide sold to someone I have never examined has no history, no medication list, and no way to change course next week. The bar for a document is higher than the bar for a conversation, and where the evidence does not support recommending something at a distance, it stays out. That is the reason you can trust what did make it in.

Running a fair trial

Timing first. Six weeks of consistent levothyroxine administration before you judge anything else.

One at a time. Selenium is the reasonable first, if you are going to trial anything.

Twelve weeks. The antibody studies ran three months; the probiotic study ran a comparable course.

Decide in advance what success means, and do not make it the antibody number. Energy, cognition, sleep, hair, cold tolerance, and how you feel by mid-afternoon are the outcomes worth tracking.

Track it: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=hashimotos-supplement

Where to find quality versions of these

Everything above was written without reference to any product line.

Third-party testing first. NSF Certified for Sport and USP Verified mean an independent lab confirmed contents. **For this condition specifically, also read the label for iodine**, since thyroid-support blends frequently include it and that is the one ingredient you most want to control.

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of sales through that link. Nothing here was included, excluded, ranked, or dosed based on what it earns. The single most-sold category in this condition, proprietary thyroid support blends, is in the do-not-buy section above.

When to seek clinical review

Urgently: chest pain, resting heart rate persistently over 120, severe confusion or drowsiness, or difficulty swallowing or breathing.

Promptly: palpitations, tremor, heat intolerance, or unintended weight loss, which can indicate over-replacement.

Before starting anything here: if you are pregnant, planning pregnancy, or taking multiple medications where timing becomes complicated.

Working with Root Health

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=hashimotos-supplement

Consultations are available to people located in North Carolina, where I am licensed.

Iron, which is easy to miss and common in this population

Thyroid peroxidase is an iron-dependent enzyme, so iron status is not a side issue in thyroid function.

A systematic review and meta-analysis of iron deficiency and thyroid function found thyroid hormone levels lower in people with iron deficiency, particularly in pregnant women, and reported a possible relationship between iron deficiency, thyroid function, and autoimmunity in some patient groups (PMID 38004184). A separate systematic review and meta-analysis in reproductive-age and pregnant women found significantly increased odds of overt and subclinical hypothyroidism in pregnant women with iron deficiency (PMID 33716980).

What to do. Test before supplementing. Ferritin with serum iron, TIBC, transferrin saturation, and a CBC. Iron is not a supplement to take speculatively: unnecessary iron causes constipation and nausea, overload is a real risk, and it is a leading cause of poisoning death in young children, so keep it locked away.

Timing matters if you take levothyroxine. Iron interferes with its absorption. Separate them by at least four hours and tell your prescriber you are taking it.

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Functional target ranges transcribed from the Root Health blood chemistry interpreter. Levothyroxine timing guidance reflects the product prescribing information.

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